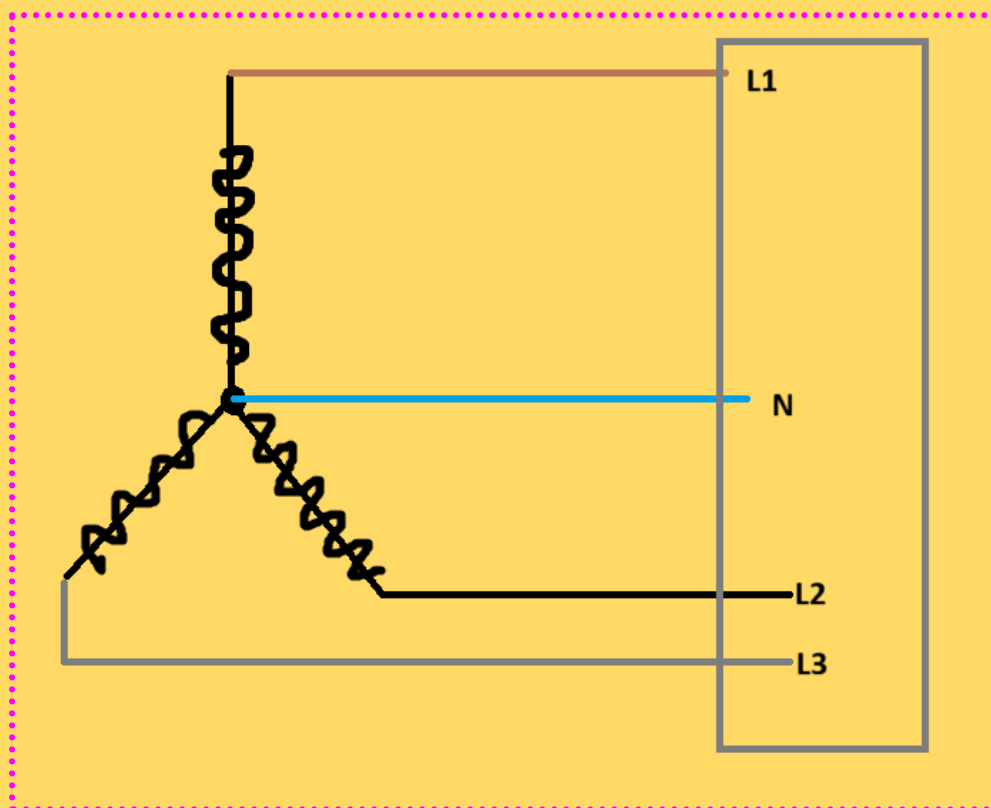


Unit 302 Mock exam

Question 10: Calculate phase voltage and current for a star wound 400V 3 phase network with 3 resistances of 16 ohms?

The star method has a neutral.



1) $I_L = I_p$

V_L the voltage line - is the voltage between the lines conductors.
 V_p the phase voltage - ?

The square root 3 is a constant.

$$V_p = \frac{V_L}{\sqrt{3}}$$

$$\frac{400}{\sqrt{3}} = 230 \text{ V}$$

230V is the Voltage in Phase. The symbol for this is V_p .

Next is the Current

$$I_L = \frac{V_p}{R}$$

$$I_L = \frac{230\text{V}}{16 \Omega}$$

$$I_L = 14.43 \text{ A}$$